

Vision-Guided Automated Foosball Table

Joonhan Ryu¹, Luke Watson¹, Daniel Yu¹, Weiqing Zhang¹ | Team 2025162

Supervisor: Dr. Raviraj Adve¹ | Administrator: Dr. Khoman Phang¹ | Section: #2

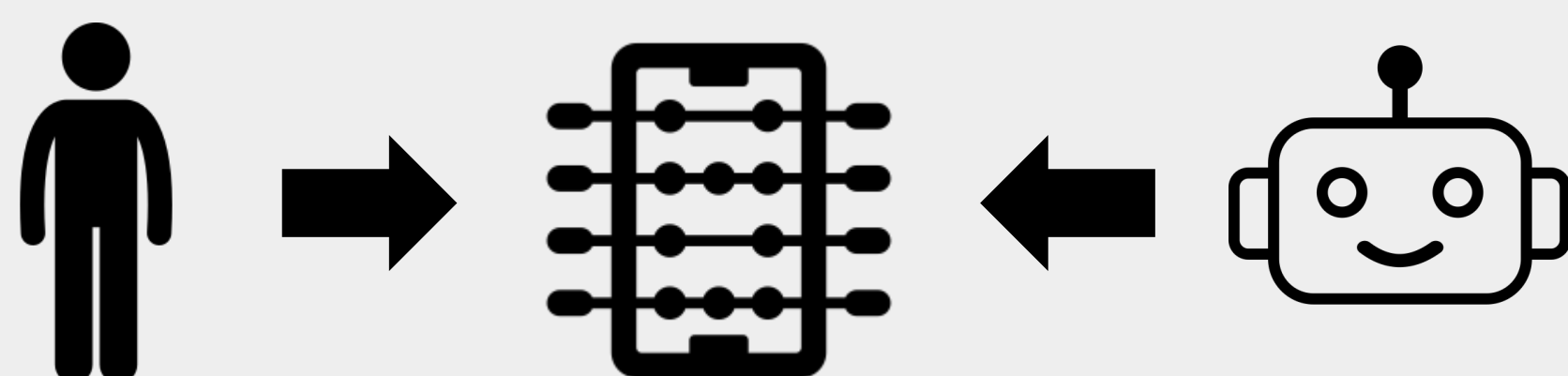
¹Faculty of Applied Science & Engineering, University of Toronto, Toronto, Canada.



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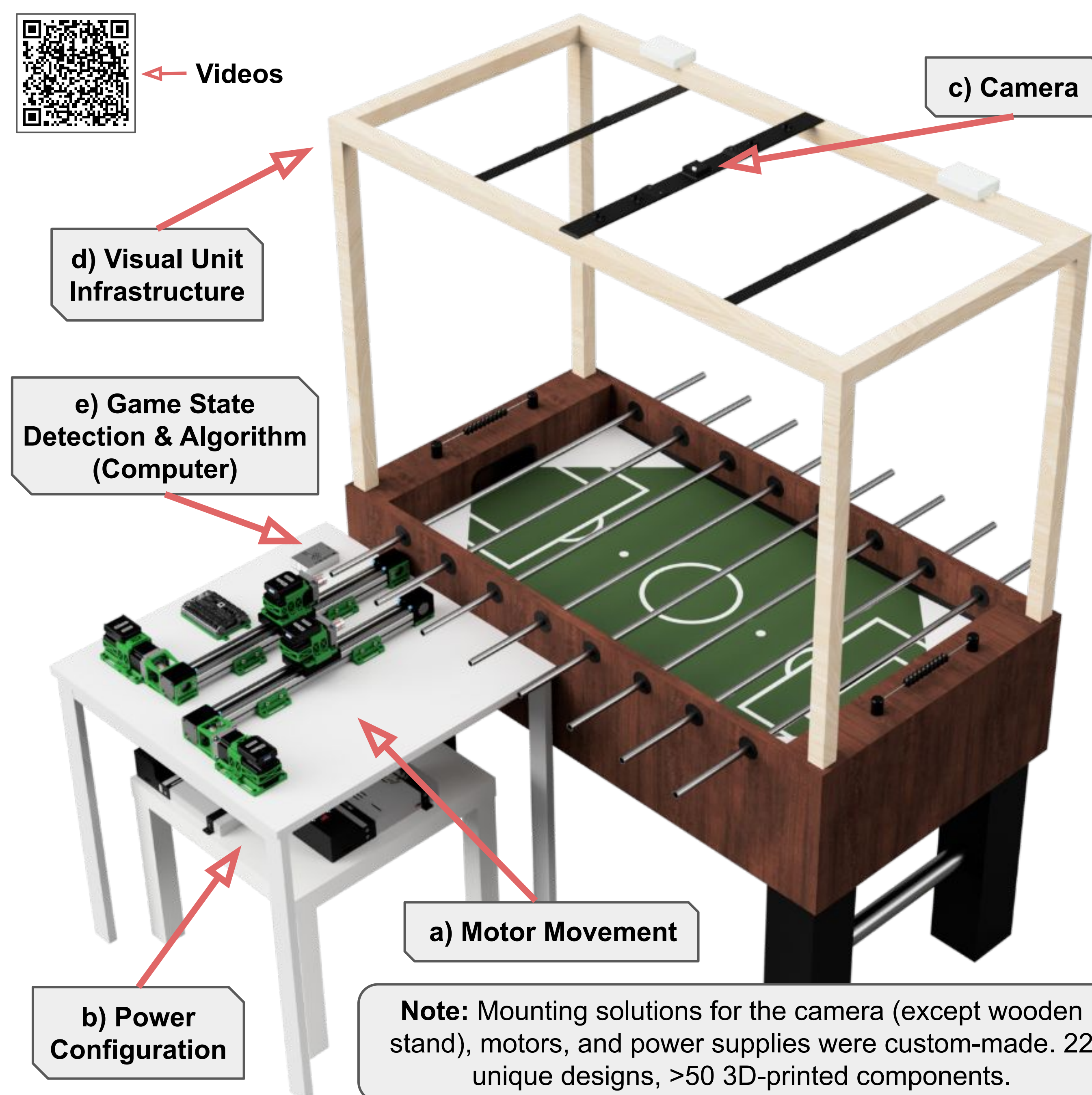
1. Background & Motivation

- Commercial solutions exist for automating a player in smaller tabletop games like Chess
- No system to automate a foosball table that can be reproduced with **off-the-shelf** components
- One project **integrates multiple areas** of engineering, including software and mechatronics



2. Project Goals & Objectives

- Automate **2 defensive foosball rods** for an interactive one-player experience
 - System can **prevent goals** for an average of **1 minute**
- Removable **accessory** to table
- Uses widely available components

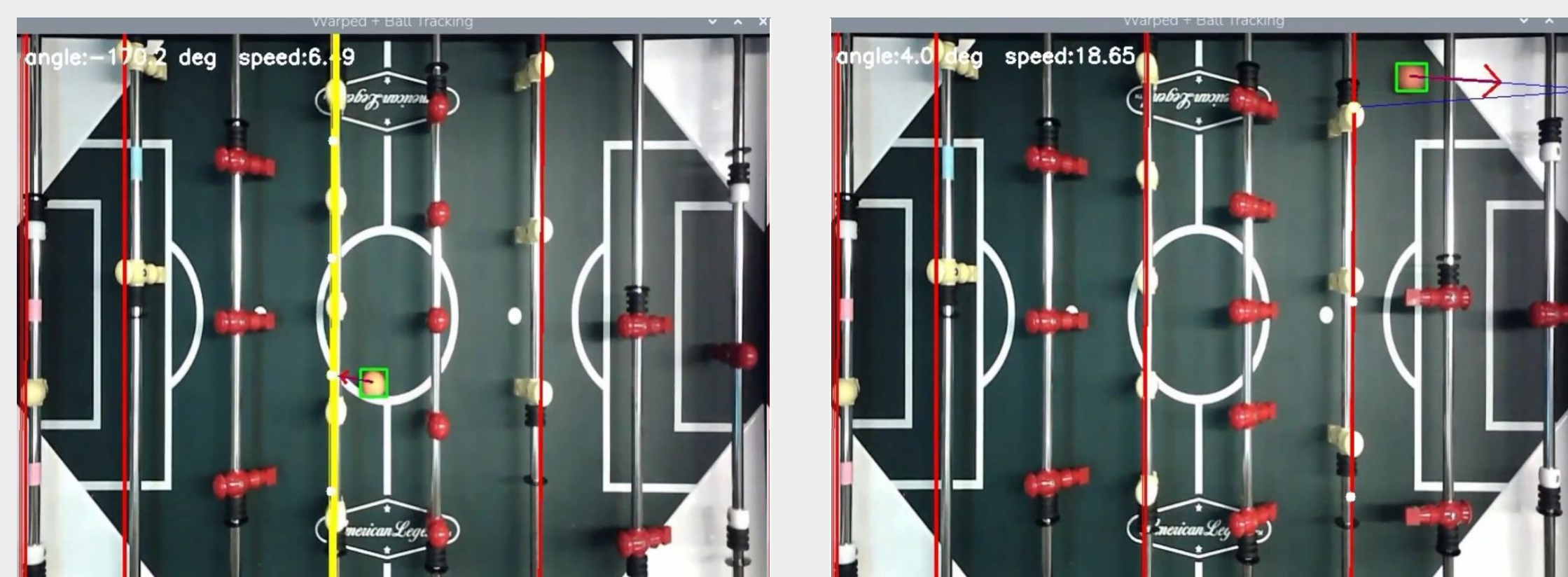


3. Requirements & Challenges

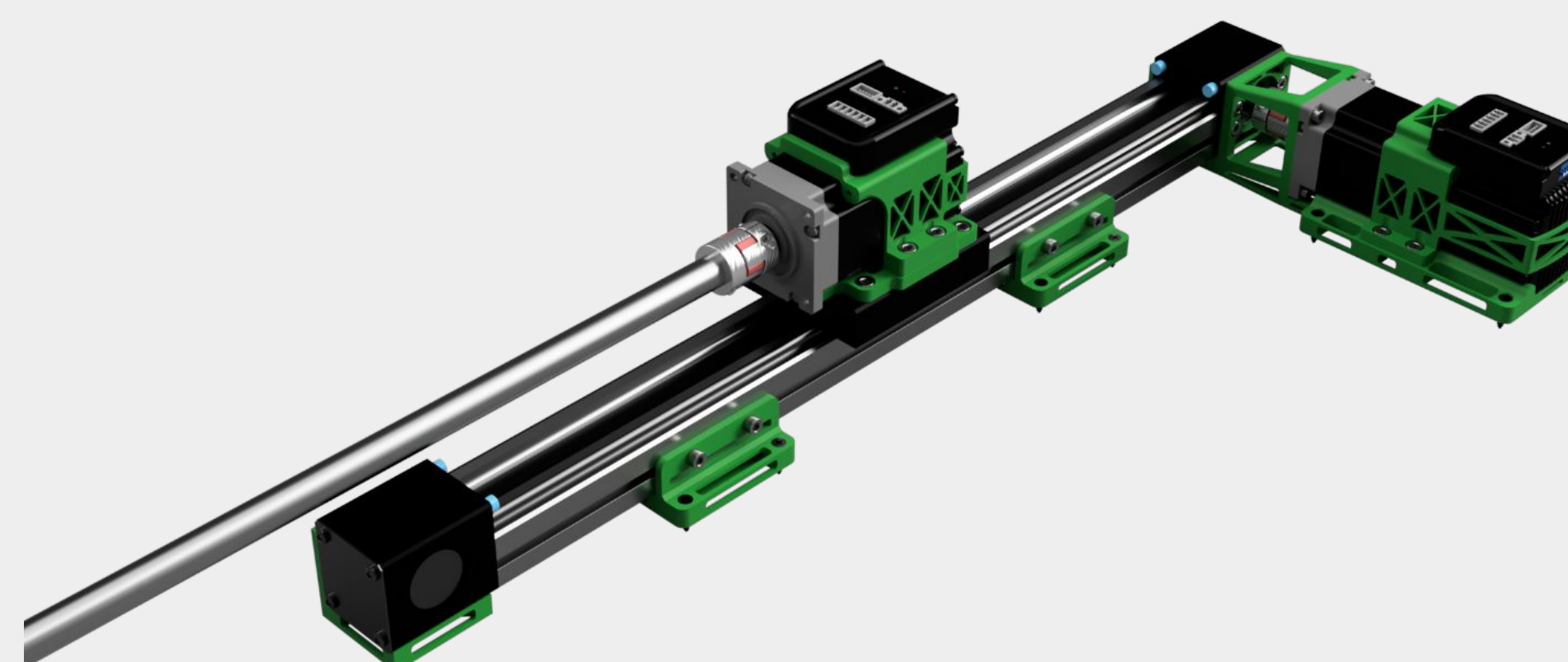
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|----------------------------------|--|
| <p>a) </p> | <ul style="list-style-type: none"> Motors rotate rods 360°, move 400mm linearly, controller converts commands to voltage signals Challenge: creating realistic speed |
| <p>b) </p> | <ul style="list-style-type: none"> Converts AC-to-DC power, delivers 36V to motors, 24V to controller Enclosures and inlet for safety Challenge: high-voltage wiring |
| <p>c) </p> <p>Arducam OV9782</p> | <ul style="list-style-type: none"> Captures field in > 60 frames per second in full colour through USB Simultaneous input (global shutter) Challenge: stream backend setup |
| <p>d) </p> | <ul style="list-style-type: none"> Supports camera, provides lighting Tall enough for entire game state to be in frame unobstructed Challenge: assembly hardware |
| <p>e) </p> <p>Raspberry Pi 5</p> | <ul style="list-style-type: none"> Detects ball & foosmen positions Calculates required movement Sends commands to motors Challenge: latency, motor overshoot |

4. Highlights & Results

- Fully automated** 2 defensive rods
- Successfully **combined high-speed camera** input with **mechanical rod actuation** for **real-time gameplay** response
- Functional pipeline: **vision** → **decision** → **actuation**
- Fast, stable ball detection via **colour filtering** and precise ball **trajectory estimation**



- Maintains **active defense** for **≥1 min** on average
- Movement system capable of stopping and kicking ball
- Controller processes software commands** and completes **motor movement** in **~1ms**
- High precision** 3D-printed mounts securely house components for safe operation



5. Conclusion & Future Work

- Efficient **colour-tracking** can replace deep-learning architectures in **game state detection**
- 3D-printed** components can **affordably integrate** industrial power supplies & motors to consumer products like foosball tables
- Algorithmic tuning** is essential to manage physical risks such as **motor overshoot**

Potential Future Improvements:

- Expanding automation to **all four rods**
- Improving **granularity** of motor control
- Revising algorithm for **offensive strategy**
- Optimizing max frame rate on hardware
- Explore potential **Deep Learning methods**